

ORIGINAL RESEARCH

Sacubitril-Valsartan vs Enalapril in Heart Failure Due to Chagas Disease



Primary Results of ANSWER-HF Randomized Trial

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ABSTRACT

BACKGROUND Chronic Chagas cardiomyopathy (CCC) represents a common cause of nonischemic heart failure with reduced ejection fraction (HFrEF) in Latin America, yet evidence from randomized trials remains scarce. Sacubitril-valsartan has shown benefits in HFrEF of other etiologies, but less is known about its effects on CCC.

OBJECTIVES The authors sought to compare the effects of sacubitril-valsartan vs enalapril on cardiac remodeling, functional outcomes, biomarkers, and safety in patients with CCC and HFrEF.

METHODS ANSWER-HF was a randomized, double-blind, controlled trial in Brazil. A total of 190 patients with CCC and HFrEF (left ventricular ejection fraction [LVEF] <40%, NYHA functional class II-IV) were randomized in a 1:1 ratio to sacubitril-valsartan or enalapril and followed for 6 months. The primary endpoint was change in LVEF from baseline to 6 months. The win ratio method analyzed the hierarchical secondary endpoint of cardiovascular death, heart failure hospitalization, N-terminal pro-B-type natriuretic (NT-proBNP) change, and LVEF change.

RESULTS Mean age was 61 years, 40% women, and 69% Black or mixed race. Baseline mean LVEF was 30.1%. At 6 months, mean LVEF increased by 2.1% with sacubitril-valsartan and 1.2% with enalapril (between-group difference 0.9 percentage points; 95% CI: -0.9 to 2.6; $P = 0.36$). In hierarchical analysis, sacubitril-valsartan achieved more wins than enalapril (win ratio 1.80; 95% CI: 1.27-2.63). Median NT-proBNP was significantly lower with sacubitril-valsartan (geometric mean ratio 0.68; 95% CI: 0.57-0.81; $P < 0.001$). No significant differences were observed in echocardiographic remodeling or 6-minute walk distance, and the occurrence of safety outcomes were comparable.

CONCLUSIONS In patients with CCC and HFrEF, sacubitril-valsartan did not result in significant improvements in LVEF compared with enalapril after 6 months, but was associated with a greater reduction in NT-proBNP. No safety concerns were associated with either the sacubitril-valsartan or enalapril group. These findings establish the feasibility and safety of conducting trials in this neglected disease, demonstrate biological activity of sacubitril-valsartan and enalapril in HFrEF due to CCC, and chart the path for larger, outcome-driven studies. The ANSWER-HF represents an important step forward, beginning a new era of evidence generation for Chagas cardiomyopathy. (Angiotensin Receptor-Nephrilysin Inhibition in Chagas Cardiomyopathy With Reduced Ejection Fraction [ANSWER-HF]; [NCT04853758](https://doi.org/10.1016/j.jacc.2025.10.053)) (JACC. 2026;87:1220-1232) © 2026 by the American College of Cardiology Foundation.

Chagas disease, caused by *Trypanosoma cruzi* infection, remains one of the most neglected tropical diseases globally, with major public health implications across Latin America.^{1,2} Global migration has expanded its impact beyond endemic regions, with growing numbers of cases identified in North America, Europe, Asia, and Oceania.³ Chronic Chagas cardiomyopathy (CCC), the most severe cardiac manifestation of the disease, accounts for a substantial burden of nonischemic heart failure in Latin America and is associated with high rates of ventricular arrhythmias, thromboembolic events, and sudden cardiac death.⁴ Patients with CCC and heart failure with reduced ejection fraction (HFrEF) experience significantly worse outcomes compared with those with HFrEF of other etiologies.⁵

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The development of HFrEF in Chagas disease patients involve unique pathophysiological mechanisms, such as persistent myocarditis, autonomic dysfunction, and microvascular abnormalities.⁶⁻⁸ These features may influence both disease progression and response to treatments. However, guideline-directed medical therapy for CCC has historically been based on extrapolation from pivotal heart failure trials that excluded this population.⁹⁻¹¹ No double-blind randomized trial had directly tested modern neurohormonal blockade in CCC, despite emerging evidence underscoring the need for targeted studies in this context.^{10,12}

In this setting, a comprehensive evaluation of patients with CCC receiving contemporary treatments may help inform clinical practice and advance evidence-based care for this neglected condition. The ANSWER-HF (Sacubitril-Valsartan Versus Enalapril in Heart Failure Due to Chagas Disease) trial was therefore conceived not merely as a test of efficacy, but as a foundational step toward feasibility, safety, and biological insight. Its mechanistic and feasibility-focused design sought to assess whether the benefits of neurohormonal blockade observed in other HFrEF etiologies extend to CCC, while also evaluating

effects on cardiac remodeling, functional parameters, and biomarkers (Angiotensin Receptor-Neprilysin Inhibition in Chagas Cardiomyopathy With Reduced Ejection Fraction [ANSWER-HF]; [NCT04853758](#)).

METHODS

STUDY DESIGN AND PARTICIPANTS. The ANSWER-HF trial was an investigator-initiated, single-center, randomized, double-blind controlled clinical trial that compared sacubitril-valsartan with enalapril in patients with CCC and HFrEF. The study was conducted at the Heart Institute (InCor), University of São Paulo Medical School in São Paulo, Brazil. The protocol was approved by the local ethics committee, and all participants provided written informed consent before any study procedure. The trial adhered to the principles of the Declaration of Helsinki and Good Clinical Practice guidelines.

Eligible participants were adults aged 18 years or older with a confirmed diagnosis of Chagas disease confirmed by 2 different serologic tests and symptoms consistent with NYHA functional class II, III, or IV heart failure. Patients were required to have a left ventricular ejection fraction (LVEF) <40% on transthoracic echocardiography performed at screening. All participants were required to be on stable and optimized doses of a beta-blocker and either an angiotensin-converting enzyme inhibitor (ACEi) or an angiotensin receptor blocker (ARB) for at least 4 weeks before randomization. The use of loop diuretic agents, mineralocorticoid receptor antagonists, and sodium-glucose cotransporter 2 (SGLT2) inhibitors was permitted and recommended as optimized background therapy. Key exclusion criteria included: symptomatic hypotension, systolic blood pressure <95 mm Hg at randomization, estimated glomerular filtration rate <30 mL/min/1.73 m², or serum potassium >5.2 mmol/L. Patients with a history of angioedema or previous serious adverse reactions to ACEi or ARB therapy were also excluded ([Supplemental Table 1](#)).

ABBREVIATIONS AND ACRONYMS

ACEi = angiotensin-converting enzyme inhibitor

ARB = angiotensin receptor blocker

CCC = chronic Chagas cardiomyopathy

HFrEF = heart failure with reduced ejection fraction

LVEF = left ventricular ejection fraction

NT-proBNP = N-terminal pro-B-type natriuretic peptide

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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RANDOMIZATION, TREATMENT, AND FOLLOW-UP.

Patients were assigned in a 1:1 ratio to receive either sacubitril-valsartan or enalapril in addition to usual care for 6 months. Randomization was performed using a concealed, central, automated web-based system. Treatment initiation occurred with individualized starting doses determined by ACEi/ARB dosing at enrollment, with blinded up-titration performed during consecutive follow-up visits until the target dose was achieved ([Supplemental Figure 1](#)). The protocol consisted of 3 dose levels titrated at 2-week intervals: stage 1: enalapril 2.5 mg twice daily or sacubitril-valsartan 50 mg twice daily; stage 2: enalapril 5 mg twice daily or sacubitril-valsartan 100 mg twice daily; and stage 3 (target): enalapril 10 mg twice daily or sacubitril-valsartan 200 mg twice daily ([Supplemental Figure 2](#)). Patients with systolic blood pressure ≥ 110 mm Hg at the randomization visit were eligible to initiate treatment at stage 2 dosing. Dose escalation was based on clinical tolerance, renal function, and serum potassium levels. Patients experiencing intolerance had adjusted dose to previous tolerated based on clinical judgment with reassessment at subsequent visits. Both patients and study personnel remained blinded throughout the treatment period, and no emergency unblinding occurred.

STUDY PROCEDURES. At the screening visit, participants who fulfilled the clinical eligibility criteria underwent baseline assessments, including blood sampling for biochemical and hematologic analyses, transthoracic echocardiography, a 24-hour Holter monitoring, a 6-minute walk test as well as collection of blood samples for biomarker analysis.

Following randomization, participants were monitored every 2 weeks for medication titration and safety assessments. Clinical evaluations, including physical examination and laboratory testing, were conducted at each visit. Two additional study visits occurred at 6-week intervals, with the final visit being conducted at 6 months. At this time, participants underwent transthoracic echocardiography, 24-hour Holter monitoring, biomarker collection, and a 6-minute walk test ([Supplemental Figure 3](#)).

Echocardiographic assessment. Transthoracic echocardiography was performed using standardized acquisition and interpretation protocols in accordance with the guidelines of the American Society of Echocardiography. LVEF was calculated using the biplane Simpson's method. Additional parameters included left ventricular dimensions, volumes, and wall thickness. All echocardiographic examinations were performed by the same dedicated team

throughout the study, ensuring consistency of acquisition and analysis. The imaging team was blinded to treatment allocation. A formal coefficient of variation was not calculated; however, reproducibility was maintained by having all studies performed and interpreted within the same core team under standardized protocols.

Holter monitoring. Ambulatory electrocardiographic monitoring was conducted using 24-hour Holter devices. Parameters of interest included minimum, mean, and maximum heart rates, burden and complexity of premature ventricular contractions, and episodes of sustained or nonsustained ventricular tachycardia. Holter-based arrhythmic endpoints were collected as per protocol and will be reported separately in a prespecified subanalysis.

6-Minute walk test. The 6-minute walk test was performed in accordance with the 2002 American Thoracic Society guidelines. The total distance walked, vital signs (blood pressure and heart rate), and patient-reported perceived exertion using the Borg scale were recorded both before and after the test.

Laboratory analyses. NT-proBNP was measured at baseline and after 6 months of follow-up. Other laboratory testing included complete blood count, serum urea, and creatinine, electrolytes (sodium, potassium, calcium, and magnesium), fasting glucose, glycated hemoglobin (HbA_{1c}), and high-sensitivity C-reactive protein.

STUDY OUTCOMES. The primary outcome was the mean change in LVEF from baseline to 6 months, as assessed by transthoracic echocardiography. The secondary outcome was a hierarchical composite of: 1) time to cardiovascular death; 2) time to first hospitalization for heart failure; 3) relative changes in NT-proBNP levels from baseline to 6 months; and 4) relative change in LVEF from baseline to 6 months. All clinical outcomes were adjudicated according to prespecified definitions. The criteria used to define cardiovascular death and hospitalization for heart failure are provided in the [Supplemental Appendix](#).

Other secondary outcomes were: 6-minute walk distance, NT-proBNP levels, NYHA functional class, measures of left ventricular remodeling obtained by echocardiography (changes in systolic and diastolic diameters, systolic and diastolic volumes, interventricular septal thickness, and posterior wall thickness). Safety outcomes included the incidence of hypotension (systolic blood pressure < 95 mm Hg), hyperkalemia (serum potassium > 5.2 mmol/L), renal dysfunction (estimated glomerular filtration rate < 30 mL/min/1.73 m²), and angioedema. A

detailed description of the study outcomes is provided in [Supplemental Table 2](#).

STATISTICAL ANALYSIS. The sample size was calculated based on the expected difference in LVEF improvement between treatment arms from baseline to 6 months. We estimated that the mean increase in LVEF would be 5.2% in the sacubitril-valsartan group and 2.5% in the enalapril group, based on prior evidence from the PROVE-HF (Prospective Study of Biomarkers, Symptom Improvement, and Ventricular Remodeling During ARNI Therapy)¹³ and SOLVD (Studies of Left Ventricular Dysfunction)¹⁴ trials. Assuming an SD of 5.5%, we estimated that a sample size of 182 patients (91 in each group) would provide the study with 80% power to detect a between-group difference of 3.0 percentage points in LVEF, at a 2-sided alpha level of 0.05, and a 20% anticipated mortality and loss to follow-up.

We included data from all patients who had undergone randomization in the analyses of the primary and secondary outcomes, according to the modified intention-to-treat principle. In this trial, the modified intention-to-treat set excluded patients who died during follow-up, because the absence of a final echocardiographic assessment precluded a valid comparison between baseline and follow-up LVEF. This modification was necessary to preserve the internal consistency of the primary analysis, given that the endpoint required paired echocardiographic data. To ensure robustness and minimize potential bias, we also performed sensitivity analyses with data imputation, in which LVEF values for patients who died were imputed as their baseline values carried forward. The primary outcome was evaluated using an analysis of covariance, with baseline LVEF and treatment group as covariates.

The hierarchical secondary outcome was analyzed using the win ratio method.¹⁵ Each participant in each group was compared with every participant in the other group using the following hierarchy: 1) time to cardiovascular death; 2) time to first heart failure hospitalization; 3) relative change in NT-proBNP from baseline to the final visit; and 4) relative change in LVEF from baseline to the final visit. A “win” was assigned to the participant with the more favorable outcome at the earliest level of the hierarchy. Time-to-event outcomes were compared up to the shortest censoring time between the 2 participants in each pair, NT-proBNP wins were defined by at least a 25% greater reduction, and LVEF wins defined by an absolute difference of at least 3.0 percentage points. The unstratified unmatched win ratio was then calculated as the ratio of total wins to total

losses, excluding ties, with CIs and *P* values derived from bootstrap resampling.

Time-to-event secondary outcomes were also analyzed individually using Kaplan-Meier curves and estimates of cardiovascular death and heart failure hospitalization at 6 months reported using Cox proportional hazard models, with treatment group and research site as fixed-effect factors. HRs, 95% CIs, and 2-sided *P* values are reported.

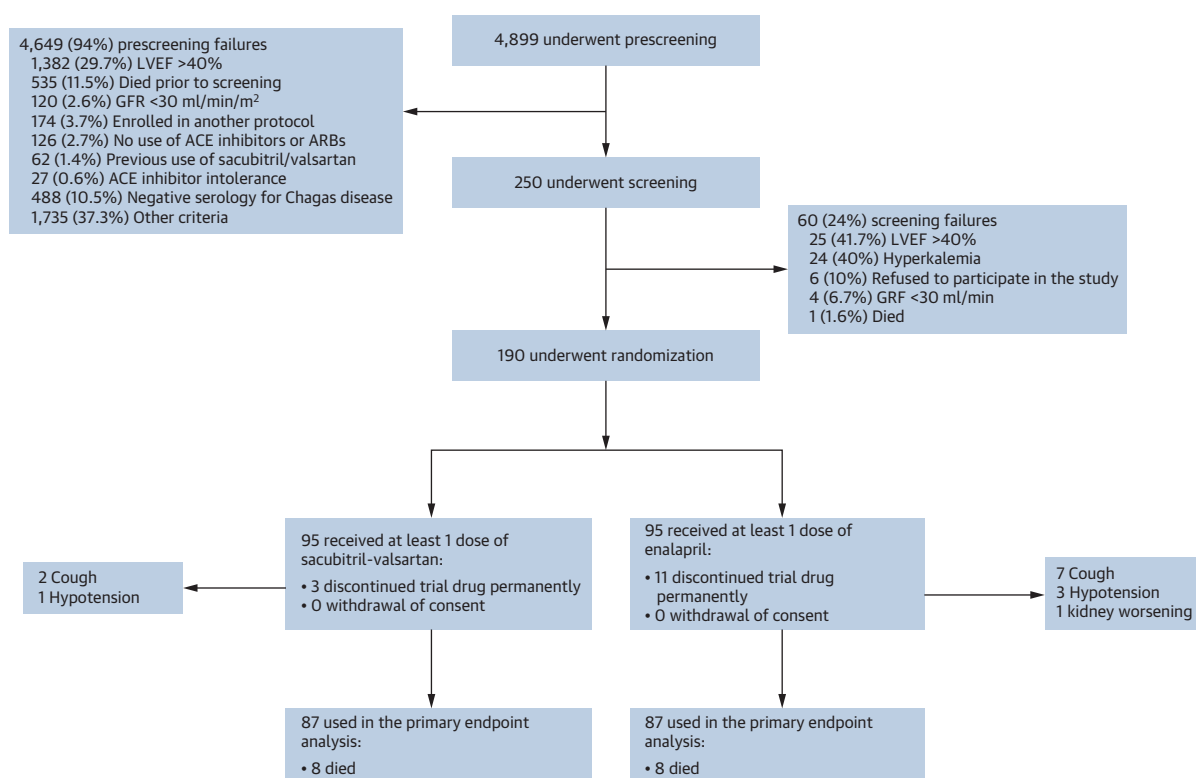
Secondary outcomes, including changes in 6-minute walk distance, NT-proBNP levels, and echocardiographic parameters of remodeling, were analyzed using analysis of covariance or generalized linear mixed models, depending on the distribution and structure of the data. Ordinal variables, such as NYHA functional class, were assessed using proportional odds models adjusted for baseline values.

The occurrence of safety outcomes was compared among study groups using Fisher exact test. All adverse events were summarized descriptively, and treatment-emergent adverse events of special interest were analyzed separately. Analyses of secondary outcomes were not adjusted for multiplicity. For the primary analysis and the win ratio analysis, patients lost to follow-up were treated as censored at the date of last contact, withdrawal of consent, or death from noncardiovascular causes. As each comparison was based on the minimum follow-up time between pairs, no imputation was performed. For continuous variables included in the hierarchical endpoint (NT-proBNP and ejection fraction), missing values were treated as ties, and isolated evaluations were performed only as complete-case analyses. All statistical analyses were conducted using R software version 4.4.1 (R Foundation for Statistical Computing). Secondary outcomes were not adjusted for multiplicity; therefore, the corresponding CIs should be interpreted with caution, and these analyses should be considered exploratory rather than definitive.

RESULTS

Between May 2021 and December 2024, a total of 4,899 individuals were prescreened ([Figure 1](#)). Of these, 4,649 were excluded, most frequently due to presenting with LVEF $\geq 40\%$, negative serology for Chagas disease, or death before screening. Ultimately, 250 individuals proceeded to the screening phase, of whom 190 met all eligibility criteria and were randomized in a 1:1 ratio to receive sacubitril-valsartan (*n* = 95) or enalapril (*n* = 95). All randomized participants initiated the allocated therapy. During follow-up, permanent discontinuation of

FIGURE 1 CONSORT Flow Diagram of the ANSWER-HF Trial



The figure depicts the flow diagram of the ANSWER-HF (Sacubitril-Valsartan Versus Enalapril in Heart Failure Due to Chagas Disease) trial. Of the 4,899 patients assessed, 190 were randomized equally to sacubitril/valsartan or enalapril. After exclusions and losses, 87 participants in each group were included in the primary endpoint analysis. ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; GFR = glomerular filtration rate; LVEF = left ventricular ejection fraction.

study medication occurred in 3 participants in the sacubitril-valsartan group and in 11 in the enalapril group, mainly attributed to adverse events. Follow-up of all participants was completed in May 2025, and database lock occurred at the end of July 2025. Over the course of follow-up, 16 participants died (8 in the sacubitril-valsartan group and 8 in the enalapril group). We had no loss of follow-up and total of 87 patients in each treatment group were included in the primary endpoint analysis, since they had complete echocardiographic information available.

BASELINE CHARACTERISTICS. Baseline demographic and clinical characteristics were well-balanced between the 2 study groups (Table 1). The mean age was 61.1 ± 9.0 years, 40.0% were women and 68.9% were Black or mixed race. At randomization, the mean systolic and diastolic blood pressures were 114.0 ± 17.1 mm Hg and 77.1 ± 58.5 mm Hg, respectively. Pharmacological therapies at randomization included prior use of angiotensin-converting enzyme

inhibitors in 71.6% of patients, angiotensin receptor blockers in 27.9%, beta-blockers in 100%, mineralocorticoid receptor antagonists in 79.5%, and SGLT2 inhibitors in 8.9%. The mean LVEF was $30.1\% \pm 6.1\%$. At baseline, median NT-proBNP concentrations were 1,564.0 pg/mL (Q1-Q3: 655.7-2,857.0 pg/mL) in the sacubitril-valsartan group and 984.0 pg/mL (Q1-Q3: 620.6-2,369.2 pg/mL) in the enalapril group. For the 6-minute walk test, the mean baseline distance was 393.5 ± 81.2 m in the sacubitril-valsartan group and 374.6 ± 79.5 m in the enalapril group.

PRIMARY OUTCOME. After 6 months of follow-up, mean LVEF increased to $32.2\% \pm 7.4\%$ in the sacubitril-valsartan group and to $31.3\% \pm 6.5\%$ in the enalapril group. The mean between-group difference was 0.9 percentage points (95% CI: -0.9 to 2.6 percentage points; $P = 0.357$). The mean LVEF values with 95% CIs are shown for each treatment group, illustrated by line plot bars (Figure 2), showed similar patterns of improvement in both treatment

arms, with overlapping interquartile ranges and comparable median values. When examining the absolute change from baseline to 6 months, both groups demonstrated a wide distribution of responses, with outliers in each treatment arm, but no statistically significant differences were observed among study groups. Sensitivity analyses with different imputation models for missing LVEF values were performed, and the results were consistent with the primary analysis. Full details are provided in [Supplemental Table 3](#).

SECONDARY OUTCOMES. In the hierarchical win ratio analysis, outcomes for time to cardiovascular death and first hospitalization for heart failure were broadly comparable. The overall signal in favor of sacubitril/valsartan was driven by the reduction in NT-proBNP at 6 months, resulting in 5,257 wins (58.2%) vs 2,906 wins (32.2%) for sacubitril/valsartan, with an unstratified win ratio of 1.81 (95% CI: 1.27-2.63; $P < 0.001$). Complete details of this analysis are shown in [Figure 3](#). When analyses were restricted to clinical outcomes, such as cardiovascular death and heart failure hospitalization at 6 months, no significant differences between groups were observed. These results are detailed in [Table 2](#) and in the Kaplan-Meier survival curves in [Supplemental Figure 4](#). Two hierarchical sensitivity analyses were performed: excluding NT-proBNP (win ratio 1.11; 95% CI: 0.74-1.69) ([Supplemental Table 4](#)) and restricted to clinical outcomes only (win ratio 0.64; 95% CI: 0.22-1.66) ([Supplemental Table 5](#)). Both analyses were neutral, showing no significant differences between groups.

NT-proBNP median concentrations at 6 months were 1,094.6 pg/mL (Q1-Q3: 437.0-2,533.5 pg/mL) in the sacubitril-valsartan group and 1091.4 pg/mL (Q1-Q3: 596.2-2,228.0 pg/mL) in the enalapril group. The geometric mean ratio for the comparison between groups was 0.68 (95% CI: 0.57-0.81; $P < 0.001$). The distribution of relative changes in NT-proBNP from baseline to 6 months is shown in the waterfall plots ([Figure 4](#)).

At 6 months, improvements in NYHA functional class were observed in both groups ([Figure 5](#)). For the 6-minute walk test, at 6 months, no significant differences were seen between groups (396.9 ± 87.3 m vs 392.5 ± 67.2 m; mean adjusted difference -8.1 m [95% CI: -23.6 to 7.4 m]; $P = 0.305$) ([Figure 6](#)). Cardiac remodeling based on echocardiographic parameters did not show any statistically significant differences between groups ([Table 3](#)).

SAFETY OUTCOMES. During follow-up, serious adverse events occurred in 16 participants (16.8%) in

TABLE 1 Baseline Characteristics

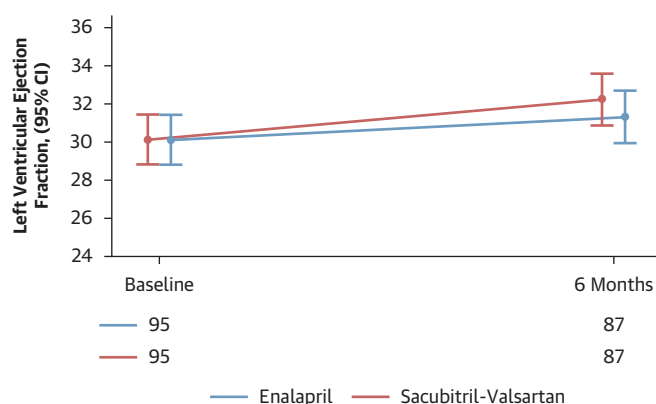
	Sacubitril-Valsartan (n = 95)	Enalapril (n = 95)	All Patients (N = 190)
Age, y	62.4 ± 8.4	59.9 ± 9.4	61.1 ± 9.0
Female	39 (41.1)	37 (38.9)	76 (40.0)
Race/ethnicity ^a			
White	31 (32.6)	27 (28.4)	58 (30.5)
Mixed race	37 (38.9)	43 (45.3)	80 (42.1)
Black	27 (28.4)	24 (25.3)	51 (26.8)
Asian	0 (0.0)	1 (1.1)	1 (0.5)
Body mass index, kg/m ² ^b	25.9 ± 4.5	26.4 ± 5.4	26.2 ± 5.0
Diastolic blood pressure, mm Hg	73.0 ± 11.3	81.1 ± 8.20	77.1 ± 5.85
Systolic blood pressure, mm Hg	115.2 ± 17.1	112.8 ± 17.1	114.0 ± 17.1
Heart rate, beats/min	65.9 ± 10.1	64.0 ± 9.5	64.9 ± 9.8
Left ventricular ejection fraction, %	30.1 ± 6.3	30.1 ± 5.8	30.1 ± 6.1
NYHA classification			
I	0 (0.0)	0 (0.0)	0 (0.0)
II	70 (73.7)	76 (80.0)	146 (76.8)
III	23 (24.2)	17 (17.9)	40 (21.1)
IV	2 (2.1)	2 (2.1)	4 (2.1)
Median NT-proBNP, pg/mL	1,564.0 (655.7-2,857.0)	984.0 (620.6-2,369.2)	1,196.0 (639.2-2,626.5)
Medical history			
Stroke	14 (14.7)	9 (9.5)	23 (12.1)
Diabetes	10 (10.5)	15 (15.8)	25 (13.2)
Dyslipidemia	31 (32.6)	27 (28.4)	58 (30.5)
Chronic obstructive pulmonary disease	2 (2.1)	1 (1.1)	3 (1.6)
Atrial fibrillation	19 (20.0)	13 (13.7)	32 (16.8)
Hypertension	49 (51.6)	30 (31.6)	79 (41.6)
Myocardial infarction	3 (3.2)	4 (4.2)	7 (3.7)
Current smoker	5 (5.3)	4 (4.2)	9 (4.7)
Previous smoker	29 (30.5)	23 (24.2)	52 (27.4)
Treatments at randomization			
Angiotensin-converting enzyme inhibitor	74 (77.9)	62 (65.3)	136 (71.6)
Angiotensin receptor blocker	21 (22.1)	32 (33.7)	53 (27.9)
Beta-blocker	95 (100)	95 (100)	190 (100)
Mineralocorticoid receptor antagonist	79 (83.2)	72 (75.8)	151 (79.5)
Loop diuretic agents	54 (56.8)	49 (51.6)	103 (54.2)
SGLT2 inhibitors	8 (8.4)	9 (9.5)	17 (8.9)
Aspirin	7 (7.4)	8 (8.4)	15 (7.9)
Antiarrhythmic agents	26 (27.4)	32 (33.7)	58 (30.5)
Anticoagulant agents	41 (43.2)	36 (37.9)	77 (40.5)
Device therapy			
Implanted cardioverter-defibrillator	21 (22.1)	19 (20)	40 (21)
Cardiac resynchronization therapy	4 (4.2)	7 (7.3)	11 (5.8)
Pacemaker	26 (27.3)	28 (29.4)	54 (28.4)

Values are mean ± SD, n (%), or median (Q1-Q3). ^aRace was determined by the patient. ^bBody mass index is the weight in kilograms divided by the square of the height in meters.

NT-proBNP = N-terminal pro-B-type natriuretic peptide; SGLT2 = sodium-glucose cotransporter 2.

each treatment group. The incidence of cough was 8.4% in the sacubitril-valsartan group and 10.5% in the enalapril group. Symptomatic hypotension was reported in 12.6% of participants in both groups.

FIGURE 2 Changes in LVEF From Baseline to 6 Months by Treatment Group



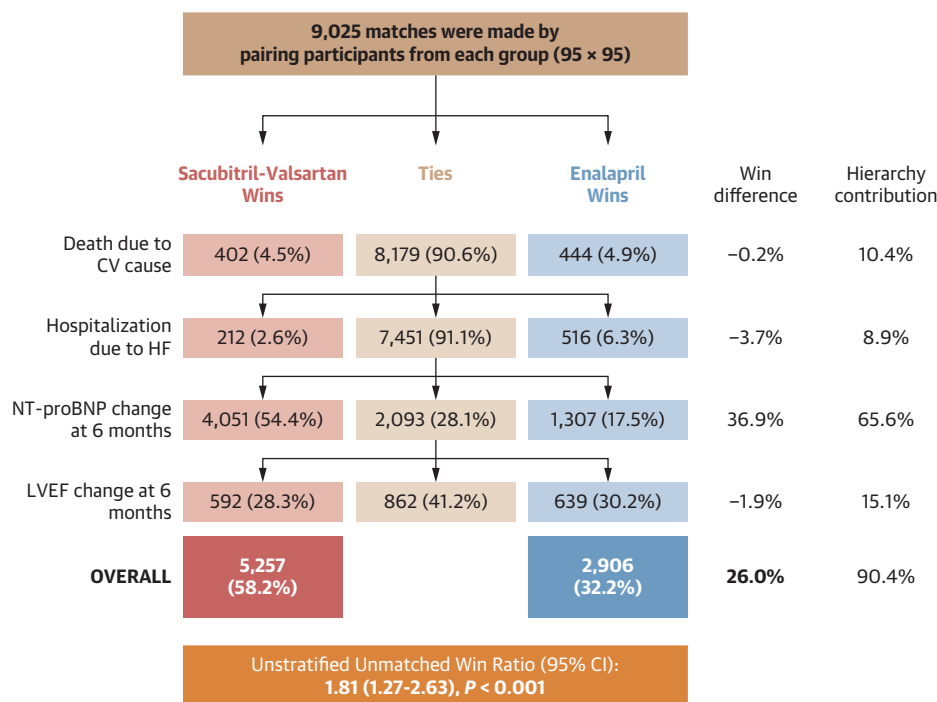
The figure shows changes in left ventricular ejection fraction (LVEF) from baseline to 6 months in patients treated with sacubitril-valsartan or enalapril. Both groups demonstrated similar trajectories, with no significant differences between therapies in either absolute LVEF values or changes from baseline.

Hyperkalemia occurred in 4.2% of patients treated with sacubitril-valsartan and 1.1% of those treated with enalapril, while renal dysfunction was reported in 3.2% and 4.2%, respectively. No meaningful differences were observed between treatment arms for any adverse event of special interest (Table 4).

DISCUSSION

In this first double-blind, randomized, controlled trial conducted in patients with CCC and HFrEF, ANSWER-HF provides evidence directly addressing a knowledge gap. Patients with this neglected disease remain largely absent from contemporary heart failure trials, leaving clinicians with little guidance on the applicability of established therapies to this high-risk population. In this context, the trial showed that sacubitril-valsartan did not result in significant improvements in LVEF or other echocardiographic parameters of cardiac remodeling compared with enalapril after 6 months, but was associated with

FIGURE 3 Hierarchical Win Ratio Analysis of Sacubitril-Valsartan vs Enalapril



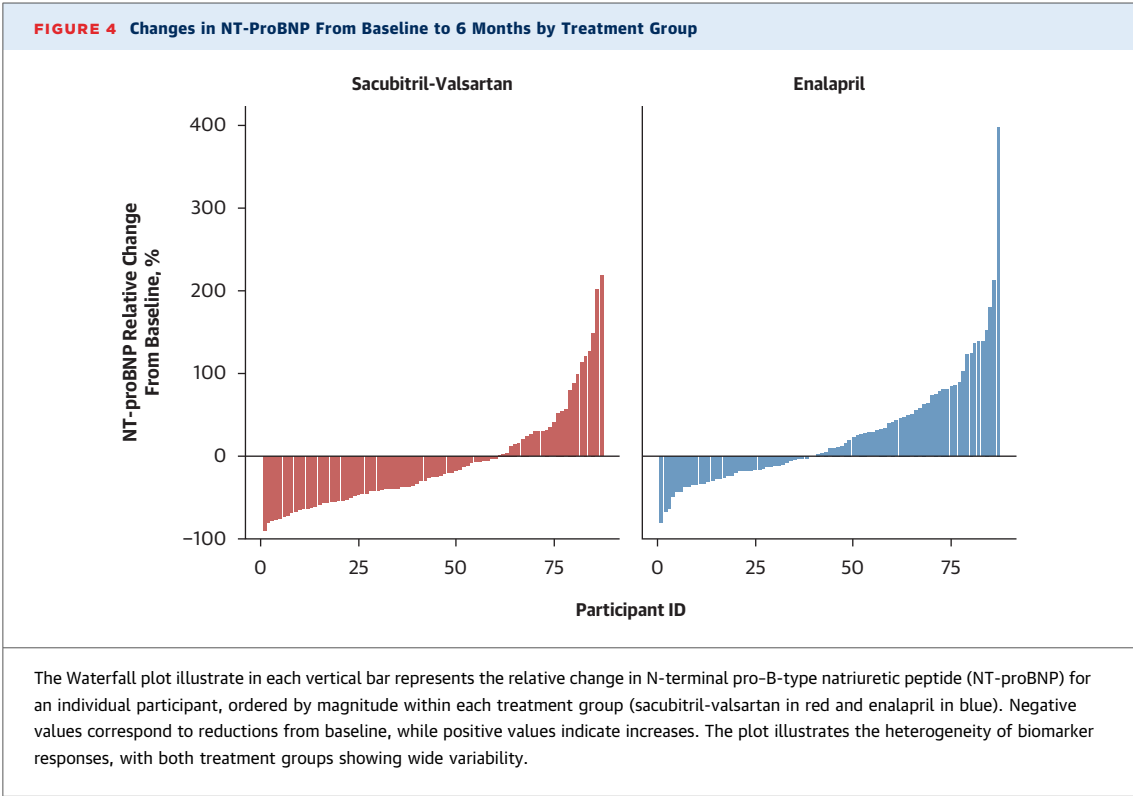
The figure summarizes the hierarchical win ratio analysis comparing sacubitril-valsartan with enalapril. A total of 9,025 matched pairs were evaluated across 4 ranked outcomes: cardiovascular (CV) death, hospitalization due to heart failure (HF), N-terminal pro-B-type natriuretic peptide (NT-proBNP) change at 6 months, and left ventricular ejection fraction (LVEF) change at 6 months. Sacubitril-valsartan achieved more wins overall (58.2%) compared with enalapril (32.2%), primarily driven by greater reductions in NT-proBNP. The unstratified unmatched win ratio was 1.81 (95% CI: 1.27-2.63; P < 0.001), favoring sacubitril/valsartan.

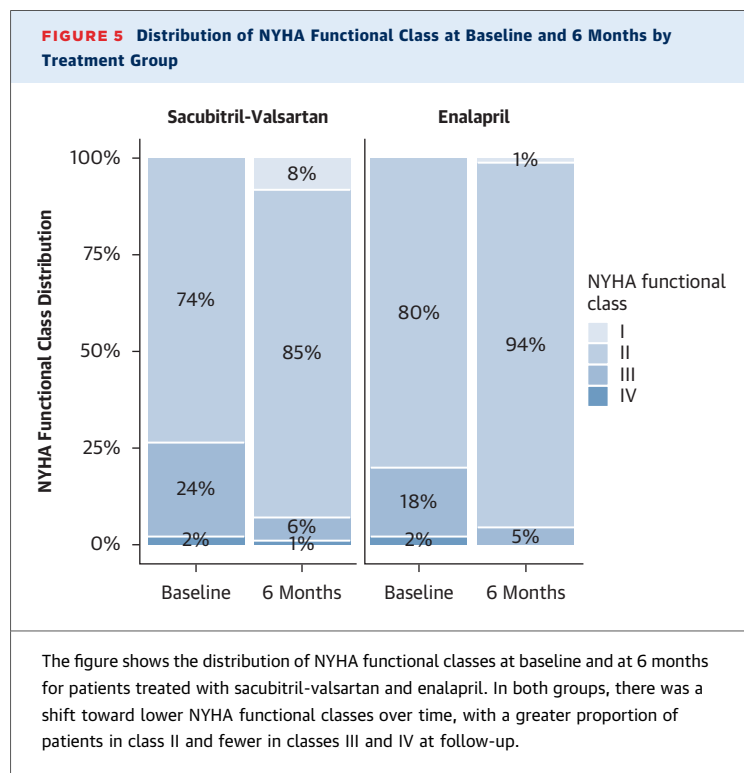
TABLE 2 Clinical Endpoints at 6 Months				
Outcomes	Sacubitril-Valsartan	Enalapril	Effect HR (95% CI)	P Value
Cardiovascular death	7/95 (7.4)	6/95 (6.3)	1.27 (0.43-3.81)	0.665
Heart failure hospitalization	7/95 (7.4)	5/95 (5.3)	1.42 (0.45-4.46)	0.553
Cardiovascular death and heart failure hospitalization	12/95 (12.6)	8/95 (8.4)	1.83 (0.72-4.67)	0.205
All-cause death	8/95 (8.4)	8/95 (8.4)	1.08 (0.40-2.89)	0.879
Values are n/N (%) unless otherwise indicated.				

greater reductions in NT-proBNP. Both treatment strategies demonstrated a consistent safety profile within this patient population. These findings highlight the biological activity of sacubitril-valsartan in Chagas cardiomyopathy, but their clinical implications remain uncertain and require confirmation in larger, adequately powered, and longer-term studies. The [Central Illustration](#) summarizes the key findings described in this paper.

Sacubitril-valsartan was tested against enalapril in prior trials involving patients with HFrEF. PARADIGM-HF (Prospective Comparison of ARNI With ACEi to Determine Impact on Global Mortality and Morbidity in Heart Failure)¹⁶ assessed chronic symptomatic patients. Subsequently, the

PARACHUTE-HF (Sacubitril/Valsartan Versus Enalapril in Chronic Chagas Cardiomyopathy)^{10,12,17} trial was conducted exclusively in patients with CCC and HFrEF. Importantly, the effects of sacubitril-valsartan vs enalapril in cardiac remodeling had not been previously explored in a prospective setting. Some similarities in patient profiles of ANSWER-HF, PARACHUTE-HF, and PARADIGM-HF may be noted, especially in terms of age, gender, and baseline LVEF. Functional status, however, differed between these 3 studies: whereas 77% of patients in ANSWER-HF were in NYHA functional class II and only 23% in functional class III to IV, PARACHUTE-HF and PARADIGM-HF enrolled a more symptomatic cohort, with a third of study





participants being in advanced classes. Median baseline NT-proBNP was also lower in ANSWER-HF compared with these 2 other trials.

In ANSWER-HF, sacubitril-valsartan effects on cardiac remodeling were comparable to those seen

with enalapril. Improvements in LVEF and additional echocardiographic parameters from baseline to 6 months were similar between study groups. This lack of structural change may reflect the biological characteristics of CCC, in which ventricular recovery may require longer follow-up or may be limited by extensive fixed fibrosis.^{3,7} These results are in contrast to the observations from the PROVE-HF study,¹⁸ which described progressive increases in LVEF at 6 months at 12 months of treatment with sacubitril-valsartan, as well as significant reductions in ventricular volumes and improved diastolic function. The EVALUATE-HF (Study of Effects of Sacubitril/Valsartan vs. Enalapril on Aortic Stiffness in Patients With Mild to Moderate HF With Reduced Ejection Fraction) trial,¹⁹ despite its shorter duration (12 weeks), also showed favorable effects of sacubitril-valsartan on ventricular and atrial volumes compared with enalapril. These differences could be explained, at least in part, by the fact the CCC, a distinct etiology of HFrEF, is often associated with more advanced structural disease and maintenance of the inflammatory pathways including the parasite persistence, which might lead to different patterns of reverse remodeling.^{4,7}

The secondary win ratio analysis favored sacubitril-valsartan over enalapril and was largely driven by greater reductions in NT-proBNP levels at 6 months in patients assigned to sacubitril-valsartan. When the analysis was restricted to

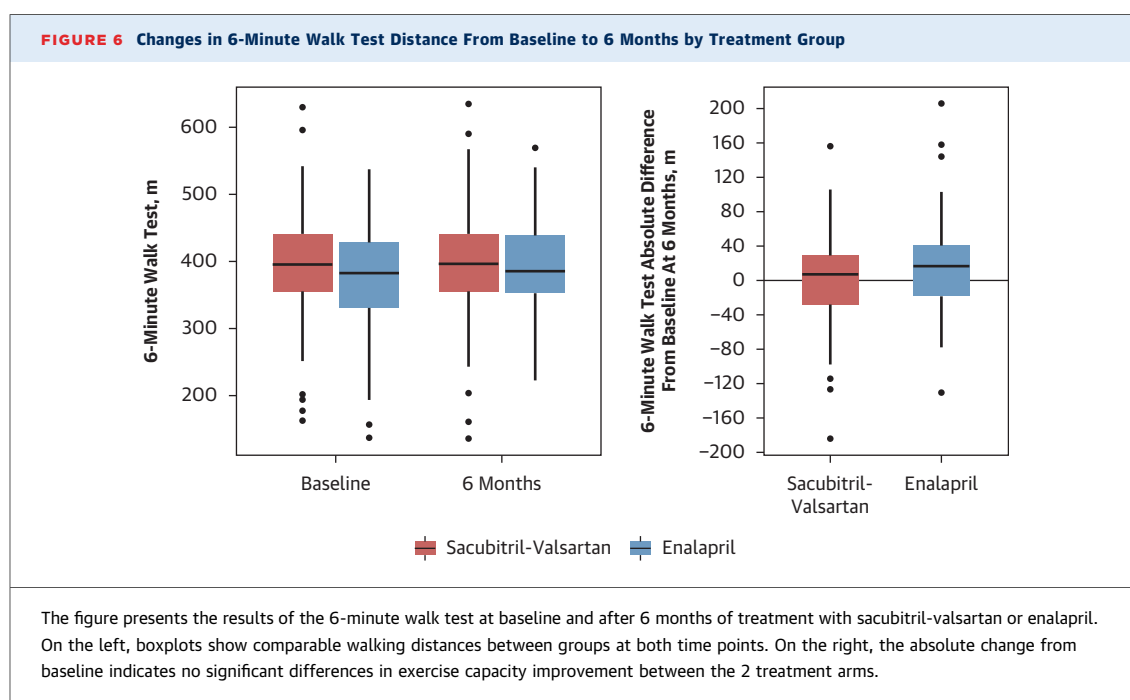


TABLE 3 Primary and Secondary Outcomes at 6 Months

Outcomes	Sacubitril-Valsartan	Enalapril	Effect MD (95% CI)	P Value
Primary endpoint				
Left ventricular ejection fraction, %	32.2 ± 7.4 (87)	31.3 ± 6.5 (87)	0.8 (−0.9 to 2.6)	0.357
Secondary endpoints				
Left ventricular end-systolic diameter, mm	54.9 ± 11.3 (87)	56.0 ± 8.2 (87)	0.3 (−1.5 to 2.1)	0.736
Left ventricular end-diastolic diameter, mm	63.7 ± 9.2 (87)	64.6 ± 6.8 (87)	−0.5 (−1.8 to 0.9)	0.496
Left ventricular end-systolic volume, mL	142.7 ± 70.3 (87)	141.1 ± 54.4 (87)	5.1 (−9.4 to 19.6)	0.490
Left ventricular end-diastolic volume, mL	201.5 ± 84.0 (87)	202.9 ± 65.0 (87)	5.0 (−12.7 to 22.7)	0.577
Interventricular septum thickness, mm	7.4 ± 1.5 (87)	7.4 ± 1.4 (87)	−0.1 (−0.5 to 0.3)	0.512
Posterior wall thickness, mm	7.0 ± 1.2 (87)	7.0 ± 1.2 (87)	−0.2 (−0.5 to 0.2)	0.347
6-minute walk distance, m	396.9 ± 87.3 (83)	392.5 ± 67.2 (87)	−8.1 (−23.6 to 7.4)	0.305
NT-proBNP, pg/mL	1,094.6 (437.0-2,533.5) (87)	1,091.4 (596.2-2,228.0) (87)	0.68* (0.57-0.81)	<0.001

Values are mean ± SD (n) or median (Q1-Q3) (n), unless otherwise indicated. *Geometric mean ratio is shown instead of mean difference (MD).
LVEF = left ventricular ejection fraction; NT-proBNP = N-terminal pro-B-type natriuretic peptide.

clinical outcomes such as cardiovascular death and heart failure hospitalization, no significant differences between groups were observed, underscoring that the overall signal was primarily biomarker-driven. This reduction in NT-proBNP may reflect neurohormonal effects independent of structural remodeling, including reduced wall stress, autonomic modulation, and attenuation of persistent inflammation.^{15,18,20}

These findings closely resemble those observed in the PARACHUTE-HF trial,^{12,15} in which sacubitril-valsartan was associated with a 52% higher likelihood of a favorable primary outcome compared with enalapril (win ratio 1.52; 95% CI: 1.28-1.82), again largely explained by early and more pronounced reductions in NT-proBNP at 3 months, whereas event rates for cardiovascular death and hospitalization remained similar between groups. We observed that in patients receiving sacubitril-valsartan, the mean NT-proBNP level was 32% lower than in those receiving enalapril. In the PIONEER-HF (Comparison of Sacubitril/Valsartan Versus Enalapril on Effect on NT-proBNP in Patients Stabilized From an Acute Heart Failure Episode) trial,²⁰ sacubitril-valsartan led to a rapid and significant 46.7% reduction in NT-proBNP compared with 25.3% for enalapril, which occurred within the first week of treatment and was sustained for at least 8 weeks.²¹ Longer-term prospective data similarly showed a marked decline in median NT-proBNP over 12 months, with the greatest drop occurring early and persisting thereafter, which was correlated with improvements in functional class, symptoms, and quality of life.^{18,19} Observational studies further supported this association, consistently linking NT-proBNP lowering with a shift toward less symptomatic NYHA functional classes.^{18,19}

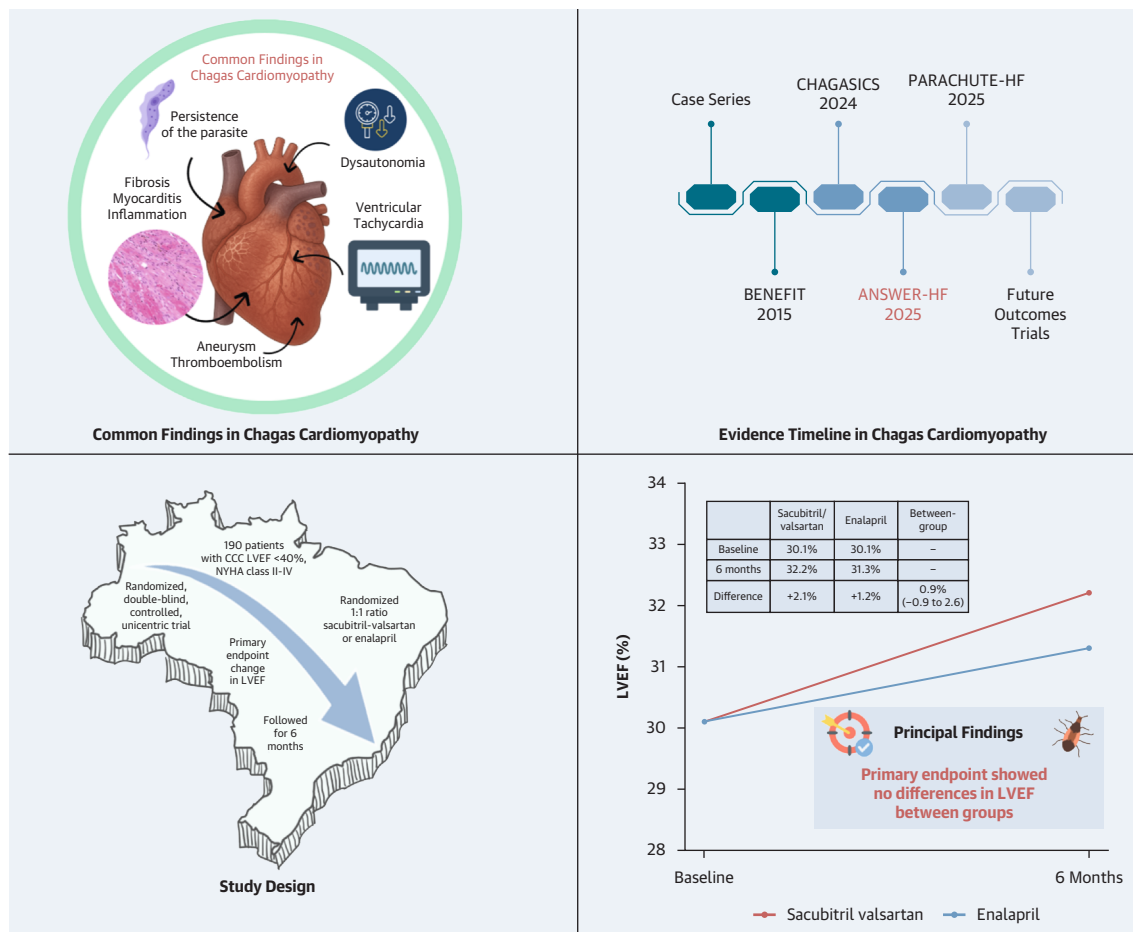
Two hierarchical sensitivity analyses were conducted to further explore the robustness of the findings. In the first, NT-proBNP was excluded, and the hierarchy was restricted to death, heart failure hospitalization, and LVEF. In the second, only clinical outcomes were considered, hierarchizing death and heart failure hospitalization. Both approaches yielded neutral results, with no significant differences between treatment groups. These observations

TABLE 4 Adverse Events Within 6 Months

Adverse Event	Sacubitril-Valsartan (n = 95)	Enalapril (n = 95)	P Value ^a
Serious adverse event			
All-cause death	8 (8.4)	8 (8.4)	1.000
Hospitalization for heart failure	7 (7.4)	5 (5.3)	0.767
Hospitalization for other causes	5 (5.3)	8 (8.4)	0.567
Sustained VT episode	0 (0)	1 (1.1)	—
Hospitalization due to pacemaker lead fracture	1 (1.1)	0 (0)	—
Hospitalization due to stroke	1 (1.1)	1 (1.1)	—
Hospitalization for ICD implantation	0 (0)	1 (1.1)	—
Hospitalization due to complete AV block with pacemaker implantation	2 (2.1)	1 (1.1)	—
Hospitalization due to sustained VT	0 (0)	1 (1.1)	—
Hospitalization for pacemaker generator replacement	0 (0)	1 (1.1)	—
Hospitalization due to acute cholecystitis	0 (0)	1 (1.1)	—
Hospitalization for cholecystectomy	0 (0)	1 (1.1)	—
Hospitalization due to acute kidney injury	1 (1.1)	0 (0)	—
Adverse events of special interest			
Cough	8 (8.4)	10 (10.5)	0.805
Symptomatic hypotension	12 (12.6)	12 (12.6)	1.000
Hyperkalemia	4 (4.2)	1 (1.1)	0.368
Renal dysfunction	3 (3.2)	4 (4.2)	1.000

Values are n (%). Values represent participants who experienced at least 1 occurrence of the specified adverse event. ^aFisher exact test.
AV = atrioventricular; ICD = implantable cardioverter-defibrillator; VT = ventricular tachycardia.

CENTRAL ILLUSTRATION Study Design and Main Results of the ANSWER-HF Trial



Madrini Jr., V, et al. JACC. 2026;87(10):1220-1232.

Integrated overview of the ANSWER-HF trial in the context of Chagas cardiomyopathy. The figure highlights common pathological findings associated with the disease, including fibrosis, myocarditis, ventricular tachycardia, aneurysm, and thromboembolism, as well as the persistence of the parasite. The study design is depicted with a randomized, double-blind, unicentric trial of 190 patients with reduced left ventricular ejection fraction (LVEF) and NYHA functional class II to IV symptoms, allocated in a 1:1 ratio to sacubitril/valsartan or enalapril, and followed for 6 months. The evidence timeline situates ANSWER-HF (Sacubitril-Valsartan Versus Enalapril in Heart Failure Due to Chagas Disease) (2025) within the broader progression of Chagas research, between prior trials such as BENEFIT (BENZnidazole Evaluation For Interrupting Trypanosomiasis) (2015),³⁰ CHAGASICS (Amiodarone Against ICD Therapy in Chagas Cardiomyopathy for Primary Prevention of Death) (2024),³¹ and PARACHUTE-HF (Sacubitril/Valsartan Versus Enalapril in Chronic Chagas Cardiomyopathy) (2025), while pointing toward future outcome studies. Finally, the principal results are summarized, showing that the primary endpoint of change in left ventricular ejection fraction did not differ between treatment groups. CCC = chronic Chagas cardiomyopathy.

reinforce that the apparent benefit identified in the full hierarchical win ratio analysis was driven almost exclusively by reductions in NT-proBNP, without evidence of improvement in clinical outcomes. Importantly, although the directionality of the analyses remained consistent, both were underpowered for definitive conclusions, suggesting that the observed signal likely reflects biological activity,

the clinical relevance of which remains to be established.

Smaller observational studies evaluating treatment with sacubitril-valsartan suggested gains in functional status by using the 6-minute walk test as a parameter.²²⁻²⁴ In ANSWER-HF, results from the 6-minute walk test showed no significant difference between sacubitril-valsartan and enalapril at

6 months. These results are consistent with the randomized OUTSTEP-HF (randOmized stUdy Using acceleromeTry to Compare Sacubitril/valsarTan and Enalapril in Patients With Heart Failure) trial,²⁵ which reported a nonsignificant between-group difference of 8.98 m. These findings indicate that whereas functional capacity may improve over time, the incremental effect of sacubitril-valsartan over ACE inhibition on the 6-minute walk test appears modest in randomized settings.

The safety profile of sacubitril-valsartan was broadly comparable to that of enalapril over 6 months, with serious adverse events occurring in 16.8% of participants in each treatment group. The incidence of cough was slightly lower with sacubitril-valsartan compared with enalapril, similar to what was observed in prior studies.^{16,26} Rates of symptomatic hypotension were identical between study arms (12.6%), contrasting with the higher incidence reported in PARADIGM-HF.²⁶ Importantly, our findings were consistent with the literature indicating no excessive renal risk with angiotensin receptor-neprilysin inhibitor therapy.²⁷⁻²⁹ Overall, the trial showed the safety profile of sacubitril-valsartan and enalapril in patients with CCC and HFrEF.

STUDY LIMITATIONS. A key limitation of the ANSWER-HF trial is its single-center design, which may restrict the external validity of the findings to other healthcare settings and patient populations. Furthermore, the relatively small sample size of 190 patients limits the statistical power to detect more nuanced differences between treatment groups and reduces the precision of effect estimates. The 6-month follow-up period, while sufficient to evaluate remodeling and biomarker changes, may not fully capture the long-term clinical impact of sacubitril-valsartan compared with enalapril in CCC. Although both sacubitril-valsartan and enalapril proved to be safe and biologically active, the study was not designed or powered to determine whether these biological effects translate into clinical outcomes, and this remains a limitation. Finally, a baseline imbalance in NT-proBNP levels between groups should be acknowledged, although this did not appear to materially affect the results and was addressed in sensitivity

analyses. These factors should be considered when interpreting the results and extrapolating them to broader populations.

CONCLUSIONS

In patients with CCC and HFrEF, sacubitril-valsartan did not result in significant improvements in LVEF compared with enalapril after 6 months, but led to a greater reduction in NT-proBNP. There were no safety concerns associated with either sacubitril-valsartan or enalapril in this population. These findings establish the feasibility and safety of conducting randomized trials in this neglected disease, demonstrate biological activity of sacubitril-valsartan and enalapril in HFrEF due to CCC and chart the path for larger, outcome-driven multicenter studies. The ANSWER-HF represents an important step forward, beginning a new era of evidence generation for CCC.

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KEY WORDS Chagas cardiomyopathy, enalapril, heart failure with reduced ejection fraction, HFrEF, sacubitril-valsartan

APPENDIX For definitions of clinical outcomes and supplemental figures and tables, as well as the study protocol and statistical analysis plan, please see the online version of this paper.